Experiment 8

AIM: Develop Inventory Management Module using ADO.NET & CRUD Operations.

Source: InventoryManagementModel

Product.cs:-

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;

namespace InventoryManagementModel.Models
{
    public class Product
    {
        public int ProductId { get; set; }
        public string ProductName { get; set; }
        public int Quantity { get; set; }
        public decimal Price { get; set; }
    }
}
```

ProductController.cs:-

```
using InventoryManagementModel.Models;
using System.Data.SqlClient;
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Mvc;
using System.Security.Cryptography.X509Certificates;

namespace InventoryManagementModel.Controllers
{
    public class ProductController : Controller
    {
        string cs =@"Data Source=(localdb)\MSSQLLocalDB;Initial
Catalog=InventoryDB;Integrated Security=True";    public ActionResult Index()
        {
            List<Product> products = new List<Product>();
            using (SqlConnection conn = new SqlConnection(cs))
            {
                string query = "SELECT * FROM Products";
                SqlCommand cmd = new SqlCommand(query, conn);
                conn.Open();
                SqlDataReader dr = cmd.ExecuteReader();
                while (dr.Read())
                {
                    products.Add(new Product
                    {
                        ProductId = Convert.ToInt32(dr["ProductId"]),
                        ProductName = dr["ProductName"].ToString(),
                        Quantity = Convert.ToInt32(dr["Quantity"]),
                        Price = Convert.ToDecimal(dr["Price"])
```

```
                });
            }
        }
        return View(products);
    }
    public ActionResult Create()
    {
        return View();
    }
    //Crete : save product
    [HttpPost]
    public ActionResult Create(Product p)
    {
        using (SqlConnection conn = new SqlConnection(cs))
        {
            string query = @"INSERT INTO Products
                    (ProductId, ProductName, Quantity, Price)
                     VALUES (@ProductId, @ProductName, @Quantity, @Price)";

            SqlCommand cmd = new SqlCommand(query, conn);

            // FIX THE TYPOS HERE (Change @ProuctId to @ProductId, etc.):
            cmd.Parameters.AddWithValue("@ProductId", p.ProductId);
            cmd.Parameters.AddWithValue("@ProductName", p.ProductName);
            cmd.Parameters.AddWithValue("@Quantity", p.Quantity);
            cmd.Parameters.AddWithValue("@Price", p.Price);

            conn.Open();
            cmd.ExecuteNonQuery();
        }
        return RedirectToAction("Index");
    }
    // 1. GET: Product/Edit/5
    [HttpGet]
    public ActionResult Edit(int id)
    {
        Product p = new Product();
        using (SqlConnection conn = new SqlConnection(cs))
        {
            string query = "SELECT * FROM Products WHERE ProductId=@ProductId";
            SqlCommand cmd = new SqlCommand(query, conn);
            cmd.Parameters.AddWithValue("@ProductId", id);
            conn.Open();
            SqlDataReader dr = cmd.ExecuteReader();
            if (dr.Read())
            {
                p.ProductId  =  Convert.ToInt32(dr["ProductId"]);
                p.ProductName = dr["ProductName"].ToString();
                p.Quantity = Convert.ToInt32(dr["Quantity"]);
                p.Price = Convert.ToDecimal(dr["Price"]);
            }
            else { return HttpNotFound(); }
        }
        return View(p);
    }

    // 2. POST: Product/Edit
    [HttpPost]
    public ActionResult Edit(Product p)
    {
```

```
            using (SqlConnection conn = new SqlConnection(cs))
            {
                string query = @"UPDATE Products SET
                        ProductName=@ProductName,
                        Quantity=@Quantity,
                        Price=@Price
                        WHERE ProductId=@ProductId";

                SqlCommand cmd = new SqlCommand(query, conn);
                cmd.Parameters.AddWithValue("@ProductId", p.ProductId);
                cmd.Parameters.AddWithValue("@ProductName", p.ProductName);
                cmd.Parameters.AddWithValue("@Quantity", p.Quantity);
                cmd.Parameters.AddWithValue("@Price", p.Price);
                conn.Open();
                cmd.ExecuteNonQuery();
            }
            return RedirectToAction("Index");
        }
        [HttpPost]
        public ActionResult Delete(int id)
        {
            using (SqlConnection conn = new SqlConnection(cs))
            {
                // FIX TYPO: Change @prouductId to @ProductId
                string query = "DELETE FROM Products WHERE ProductId=@ProductId";

                SqlCommand cmd = new SqlCommand(query, conn);

                // FIX TYPO: Change @prouctId to @ProductId
                cmd.Parameters.AddWithValue("@ProductId", id);

                conn.Open();
                cmd.ExecuteNonQuery();
            }
            return RedirectToAction("Index");
        }
    }
}
```

Create.cshtml:-

```
@model InventoryManagementModel.Models.Product
@{
    Layout = null;
}
<h2>ADD NEW PRODUCT</h2>
@using (Html.BeginForm())
{
    <p>Product Name:</p>
    @Html.TextBoxFor(m => m.ProductName)
    <br />
    <br />

    <p>Quantity:</p>
    @Html.TextBoxFor(m => m.Quantity)
    <br />
    <br />

    <p>Price:</p>
                    => m.Price)
```

```
    <br />
    <br />

    <input type="submit" value="SAVE PRODUCT" />
}
<br />
@Html.ActionLink("Back to Inventory", "Index")
```

Index.cshtml:-

```
@model List<InventoryManagementModel.Models.Product>
@{ Layout = null; }
<h2> INVENTORY MANAGEMENT </h2>
<p> @Html.ActionLink("Add new product", "Create")</p>
<table border="1">
    <tr>
        <th>Product ID</th>
        <th>Product Name</th>
        <th>Quantity</th>
        <th>Price</th>
        <th>Actions</th>
    </tr>
    @foreach (var item in Model)
    {
        <tr>
            <td>@item.ProductId</td>
            <td>@item.ProductName</td>
            <td>@item.Quantity</td>
            <td>@item.Price</td>
            <td>
                @Html.ActionLink("Edit", "Edit", new { id = item.ProductId })
                @using (Html.BeginForm("delete", "Product", new { id = item.ProductId },
                    FormMethod.Post))
                {<input type="submit" value="Delete" />}
            </td>
        </tr>
    }
</table>
```

Edit.cshtml:-

```
@model InventoryManagementModel.Models.Product
@{
    Layout = null;
}
<h2>EDIT PRODUCT</h2>
@using (Html.BeginForm())
{
    @Html.HiddenFor(m => m.ProductId)

    <p>Product Name:</p>
    @Html.TextBoxFor(m => m.ProductName)
    <br />
    <br />

    <p>Quantity:</p>
                    => m.Quantity)
```

```
    <br />
    <br />

    <p>Price:</p>
    @Html.TextBoxFor(m => m.Price)
    <br />
    <br />

    <input type="submit" value="UPDATE PRODUCT" />
}
<br />
@Html.ActionLink("Back to Inventory", "Index")
```

RoutConfig.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Mvc;
using System.Web.Routing;

namespace InventoryManagementModel
{
  public class RouteConfig
  {
    public static void RegisterRoutes(RouteCollection routes)
    {
      routes.IgnoreRoute("{resource}.axd/{*pathInfo}");

      routes.MapRoute(
        name: "Default",
        url: "{controller}/{action}/{id}",
        defaults: new { controller = "Product", action = "Index", id =
UrlParameter.Optional }
      );
    }
  }
}
```

Output:

INVENTORY MANAGEMENT

Add new product

Product ID	Product Name	Quantity	Price	Actions

INVENTORY MANAGEMENT

Add new product

Product ID	Product Name	Quantity	Price	Actions
0	Mobile	1	30000.00	Edit Delete

EDIT PRODUCT

Product Name:

Mobile

Quantity:

1

Price:

30000.00

UPDATE PRODUCT

Back to Inventory

ADD NEW PRODUCT

Product Name:

laptop

Quantity:

1

Price:

125000

SAVE PRODUCT

Back to Inventory

INVENTORY MANAGEMENT

Add new product

Product ID	Product Name	Quantity	Price	Actions
0	Mobile	1	40000.00	Edit Delete
0	laptop	1	125000.00	Edit Delete